3.5 Linking pairs of vertices

Let G be a graph, and let $X \subseteq V(G)$ be a set of vertices. We call X *linked* in G if whenever we pick distinct vertices $s_1, \dots, s_\ell$, $t_1, \dots, t_\ell$ in X we can find disjoint paths $P_1, \dots, P_\ell$ in G such that each P_i links s_i to t_i and has no inner vertex in X. Thus, unlike in Menger's theorem, we are not merely asking for disjoint paths between two *sets* of vertices: we insist that each of these paths shall link a specified pair of endvertices. *linked*

If $|G| \geqslant 2k$ and every set X of at most $2k$ vertices is linked in G, then G is *k-linked*. Clearly, this is equivalent to requiring merely that $|G| \geqslant 2k$ and disjoint paths $P_i = s_i \dots t_i$ exist for every choice of *exactly* $2k$ distinct vertices $s_1, \dots, s_k$, $t_1, \dots, t_k$: just add dummy vertices to X to bring it up to size $2k$. In practice, the latter is easier to prove, because we need not worry about inner vertices in X. *k-linked*

Clearly, every k-linked graph is k-connected. The converse, however, seems far from true: being k-linked is clearly a much stronger property than k-connectedness. Still, we shall prove in this section that we can force a graph to be k-linked by assuming that it is $f(k)$-connected, for some function $f\colon \mathbb{N} \to \mathbb{N}$. We first borrow a lemma from Chapter 7 to give a nice and simple proof that such a function f exists at all. In the remainder of the section we then prove that f can even be chosen linear.

The basic idea in the simple proof is as follows. If we can prove that G contains a subdivision K of a large complete graph, we can use Menger's theorem to link the vertices of X disjointly to branch vertices of K, and then hope to pair them up as desired through the subdivided edges of K. This requires, of course, that our paths do not hit too many of the subdivided edges before reaching the branch vertices of K.

The lemma saying that large enough connectivity does indeed force the existence of such a complete topological minor K will be proved in Chapter 7.2, where we consider several results of this type. By Theorem 1.4.3 it suffices to assume that G has large average degree:

Lemma 3.5.1. *There is a function $h\colon \mathbb{N} \to \mathbb{N}$ such that, for every $r \in \mathbb{N}$, every graph of average degree at least $h(r)$ contains K^r as a topological minor.*

Theorem 3.5.2. (Jung 1970; Larman & Mani 1970)
There is a function $f\colon \mathbb{N} \to \mathbb{N}$ such that every $f(k)$-connected graph is k-linked, for all $k \in \mathbb{N}$.

Proof. We prove the assertion for $f(k) = h(3k) + 2k$, where h is a function as in Lemma 3.5.1. Let G be an $f(k)$-connected graph. Then $d(G) \geqslant \delta(G) \geqslant \kappa(G) \geqslant h(3k)$; let K be a TK^{3k} in G as provided by Lemma 3.5.1, and let U denote its set of branch vertices. (3.3.1) G K U

For the proof that G is k-linked, let distinct vertices $s_1, \dots, s_k$ and $t_1, \dots, t_k$ be given. By definition of $f(k)$, we have $\kappa(G) \geqslant 2k$. s_i, t_i

Hence by Menger's theorem (3.3.1), G contains disjoint paths $P_1, \dots, P_k$,
P_i, Q_i $Q_1, \dots, Q_k$, such that each P_i starts in s_i, each Q_i starts in t_i, and all
$\mathcal{P}$ these paths end in U but have no inner vertices in U. Let the set $\mathcal{P}$ of these paths be chosen so that their total number of edges outside $E(K)$ is as small as possible.

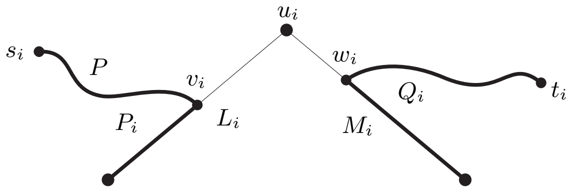


Fig. 3.5.1. Constructing an s_i–t_i path via u_i

Let $u_1, \dots, u_k$ be those k vertices in U that are not an end of a path in $\mathcal{P}$. For each $i = 1, \dots, k$, let L_i be the U-path in K (i.e., the subdivided edge of the K^{3k}) from u_i to the end of P_i in U, and let v_i be the first vertex of L_i on any path $P \in \mathcal{P}$. By definition of $\mathcal{P}$, P has no more edges outside $E(K)$ than $Pv_iL_iu_i$ does, so $v_iP = v_iL_i$ and hence $P = P_i$ (Fig. 3.5.1). Similarly, if M_i denotes the U-path in K from u_i to the end of Q_i in U, and w_i denotes the first vertex of M_i on any path in $\mathcal{P}$, then this path is Q_i. Then the paths $s_iP_iv_iL_iu_iM_iw_iQ_it_i$ are disjoint for different i and show that G is k-linked. $\square$

The function h of Lemma 3.5.1 which our proof in Chapter 7.2 will yield will be exponential in r, and will therefore give only an exponential upper bound for the function $f(k)$ in Theorem 3.5.2. As $2\varepsilon(G) \geqslant \delta(G) \geqslant \kappa(G)$, the following result implies the linear bound of $f(k) = 16k$:

[7.2.3] **Theorem 3.5.3.** (Thomas & Wollan 2005)
Let G be a graph and $k \in \mathbb{N}$. If G is $2k$-connected and $\varepsilon(G) \geqslant 8k$, then G is k-linked.

We begin our proof of Theorem 3.5.3 with a lemma.

Lemma 3.5.4. *Any graph H with $\delta(H) \geqslant 8k \geqslant |H|/2$ has a k-linked subgraph.*

Proof. If H itself is k-linked there is nothing to show, so suppose not.
X Then we can find a set X of $2k$ vertices $s_1, \dots, s_k, t_1, \dots, t_k$ that cannot
$\mathcal{P}$ be linked in H by disjoint paths $P_i = s_i \dots t_i$. Let $\mathcal{P}$ be a set of as many such paths as possible, without inner vertices in X and all of length at most 7. If there are several such sets $\mathcal{P}$, we choose one with $|\bigcup \mathcal{P}|$
$s_1,\ t_1$ minimum. We may assume that $\mathcal{P}$ contains no path from s_1 to t_1. Let

J be the subgraph of H induced by X and all the vertices on the paths in $\mathcal{P}$, and let $K := H - J$. J, K

Note that each vertex $v \in K$ has at most three neighbours on any given $P_i \in \mathcal{P}$: if it had four, then replacing the segment uP_iw between its first and its last neighbour on P_i by the path uvw would reduce $|\bigcup\mathcal{P}|$ and thus contradict our choice of $\mathcal{P}$. So v has at most 3 neighbours in J for every $i = 1, \dots, k$, at most $3k$ in total. As $\delta(H) \geqslant 8k$ by assumption, as well as $|H| \leqslant 16k$ and $|X| = 2k$, we deduce that

$$\delta(K) \geqslant 5k \quad \text{and} \quad |K| \leqslant 14k\,. \tag{1}$$

Our next aim is to show that K is disconnected. Since each of the paths in $\mathcal{P}$ has at most eight vertices, we have $|J - \{s_1, t_1\}| \leqslant 8(k-1)$. Therefore s_1 has a neighbour s in K, and t_1 has a neighbour t in K. Put $S := \{\, s' \in K \mid d_K(s,s') \leqslant 2 \,\}$ and $T := \{\, t' \in K \mid d_K(t,t') \leqslant 2 \,\}$. Since $H - \bigcup\mathcal{P}$ contains no s_1–t_1 path of length at most 7, we have $S \cap T = \emptyset$, and there is no S–T edge in K. To prove that K is disconnected, it thus suffices to show that $V(K) = S \cup T$. But for any vertex $v \in K - (S \cup T)$ the sets $N_K(s)$, $N_K(t)$ and $N_K(v)$ are disjoint and each have size at least $5k$, contradicting (1).

So K is disconnected; let C be its smaller component. By (1),

$$2\delta(C) \geqslant 2\delta(K) \geqslant 7k + 3k \geqslant \tfrac{1}{2}\,|K| + 3k \geqslant |C| + 3k\,. \tag{2}$$

We complete the proof by showing that C is k-linked. As $\delta(C) \geqslant 5k$, we have $|C| \geqslant 2k$. Let Y be a set of at most $2k$ vertices in C. By (2), every two vertices in Y have at least $3k$ common neighbours, at least k of which lie outside Y. We can therefore link any desired $\ell \leqslant k$ pairs of vertices in Y inductively by paths of length 2 whose inner vertex lies outside Y. $\square$

Before we launch into the proof of Theorem 3.5.3, let us look at its main ideas. To prove that G is k-linked, we have to consider a given set X of up to $2k$ vertices and show that X is linked in G. Ideally, we would like to use Lemma 3.5.4 to find a linked subgraph L somewhere in G, and then use our assumption of $\kappa(G) \geqslant 2k$ to obtain a set of $|X|$ disjoint X–L paths by Menger's theorem (3.3.1). Then X could be linked via these paths and L, completing the proof.

Unfortunately, we cannot expect to find a subgraph H such that $\delta(H) \geqslant 8k$ and $|H| \leqslant 16k$ (in which L could be found by Lemma 3.5.4); cf. Corollary 11.2.3. However, it is not too difficult to find a minor $H \preccurlyeq G$ that has such a subgraph (Ex. 21, Ch. 7), even so that the vertices of X come to lie in distinct branch sets of H. We may then regard X as a subset of $V(H)$, and Lemma 3.5.4 provides us with a linked subgraph L of H. The only problem now is that H need no longer be $2k$-connected, that is, our assumption of $\kappa(G) \geqslant 2k$ will not ensure that we can link X to L by $|X|$ disjoint paths in H.

And here comes the clever bit of the proof: it relaxes the assumption of $\kappa \geqslant 2k$ to a weaker assumption that does get passed on to H. This weaker assumption is that *if* we can separate X from another part of G (or H) by fewer than $|X|$ vertices, then this other part must be 'light': roughly, its own value of ε must not exceed $8k$. If X then fails to link to L by $|X|$ disjoint paths, and hence H has a separation $\{A, B\}$ with $X \subseteq A$ and $L \subseteq B$ and $|A \cap B| < |X|$, we know that ε is still at least $8k$ on $H[A]$, because the B-part of H was light.

The idea now is to continue the proof inside $H' := H[A]$ by induction. This still needs some ingenuity, since it is not enough that ε is large on H': we also need that for every low-order separation (A', B') of H' with $X \subseteq A'$ the B'-part is light. That need not be true. But when it fails, we shall be able to use induction on $H'[B']$ to show that $A' \cap B'$ is linked in $H'[B']$, and use this for our proof that X is linked in H.

X-separation

Given $k \in \mathbb{N}$, a graph G, and $A, B, X \subseteq V(G)$, call the ordered pair (A, B) an X-*separation* of G if $\{A, B\}$ is a proper separation of G of order at most $|X|$ and $X \subseteq A$. An X-separation (A, B) is *small* if $|A \cap B| < |X|$, and *linked* if $A \cap B$ is linked in $G[B]$.

small/linked

$\|\ \|^+$

light

heavy

Call a set $U \subseteq V(G)$ *light* in G if $\|U\|^+ \leqslant 8k\,|U|$, where $\|U\|^+$ denotes the number of edges of G with at least one end in U. A set of vertices is *heavy* if it is not light.

k

Proof of Theorem 3.5.3. We shall prove the following, for fixed $k \in \mathbb{N}$:

$G = (V, E)$

X

Let $G = (V, E)$ be a graph and $X \subseteq V$ a set of at most $2k$ vertices. If $V \setminus X$ is heavy and for every small X-separation (A, B) the set $B \setminus A$ is light, then X is linked in G. $(*)$

To see that $(*)$ implies the theorem, assume that $\kappa(G) \geqslant 2k$ and $\varepsilon(G) \geqslant 8k$, and let X be a set of exactly $2k$ vertices. Then G has no small X-separation. And $V \setminus X$ is heavy, since

$$\|V \setminus X\|^+ \geqslant \|G\| - \binom{2k}{2} > 8k\,|V| - 16k^2 = 8k\,|V \setminus X|\,.$$

By $(*)$, X is linked in G, completing the proof that G is k-linked.

We prove $(*)$ by induction on $|G|$, and for each value of $|G|$ by induction on $\|V \setminus X\|^+$. If $|G| = 1$ then X is linked in G. For the induction step, let G and X be given as in $(*)$. We first prove the following:

We may assume that G has no linked X-separation. (1)

(A, B)

S

For our proof of (1), suppose that G has a linked X-separation (A, B). Let us choose one with A minimal, and put $S := A \cap B$.

We first consider the case that $|S| = |X|$. If $G[A]$ contains $|X|$ disjoint X–S paths, then X is linked in G because (A, B) is linked, completing the proof of $(*)$. If not, then by Menger's theorem (3.3.1) $G[A]$ has a small X-separation (A', B') such that $B' \supseteq S$. If we choose this of minimum order, i.e. with $|A' \cap B'|$ minimum, we can link $A' \cap B'$ to S in $G[B']$ by $|A' \cap B'|$ disjoint paths, again by Menger's theorem. But then $(A', B' \cup B)$ is a linked X-separation of G that contradicts the choice of (A, B).

So $|S| < |X|$. Let G' be obtained from $G[A]$ by adding any missing edges on S, so that $G'[S]$ is a complete subgraph of G'. As (A, B) is now a small X-separation, our assumption in $(*)$ says that $B \setminus A$ is light in G. Thus, G' arises from G by deleting $|B \setminus A|$ vertices outside X and at most $8k\,|B \setminus A|$ edges, and possibly adding some edges. As $V \setminus X$ is heavy in G, this implies that G'

$$A \setminus X \textit{ is heavy in } G'.$$

In order to be able to apply the induction hypothesis to G', let us show next that for every small X-separation (A', B') of G' the set $B' \setminus A'$ is light in G'. Suppose not, and choose a counterexample (A', B') with B' minimal. As $G'[S]$ is complete, we have $S \subseteq A'$ or $S \subseteq B'$. (A', B')

If $S \subseteq A'$ then $B \cap B' \subseteq S \subseteq A'$, so $(A' \cup B, B')$ is a small X-separation of G. Moreover,

$$B' \setminus (A' \cup B) = B' \setminus A',$$

and no edge of $G' - E$ on S is incident with this set (Fig 3.5.2). Our assumption that this set is heavy in G', by the choice of (A', B'), therefore implies that it is heavy also in G. As $(A' \cup B, B')$ is a small X-separation of G, this contradicts our assumptions in $(*)$.

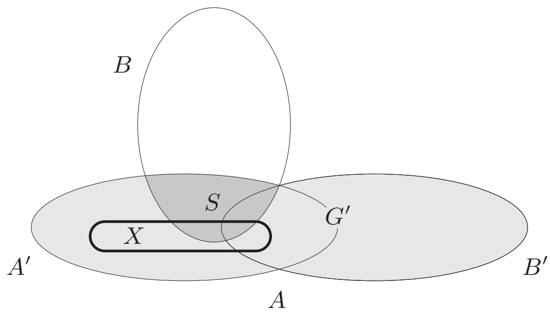


Fig. 3.5.2. If $S \subseteq A'$, then $(A' \cup B, B')$ is an X-separation of G

Hence $S \subseteq B'$. By our choice of (A', B'), the graph $G'' := G'[B']$ satisfies the premise of $(*)$ for $X'' := A' \cap B'$. Indeed, $B' \setminus X'' = B' \setminus A'$

is heavy, and by the minimality of B' any small X''-separation (A'', B'') of G'' will be such that $B'' \setminus A''$ is light, because $(A' \cup A'', B'')$ will be a small X-separation of G', and $B'' \setminus A'' = B'' \setminus (A' \cup A'')$.

By the induction hypothesis, therefore, X'' is linked in G''. But then X'' is also linked in $G[B' \cup B]$: as S was linked in $G[B]$, we simply replace any edges added on S in the definition of G' by disjoint paths through B (Fig. 3.5.3). But now $(A', B' \cup B)$ is a linked X-separation of G that violates the minimality of A in the choice of (A, B).

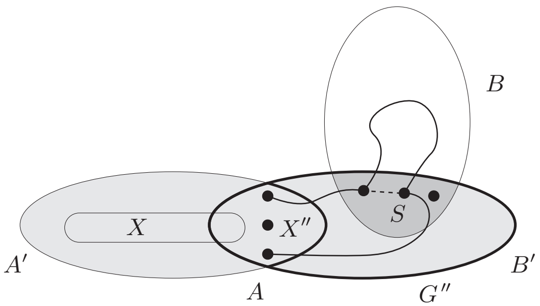


Fig. 3.5.3. If $S \subseteq B'$, then $(A', B' \cup B)$ is linked in G

We have thus shown that G' satisfies the premise of $(*)$ with respect to X. Since $\{A, B\}$ is a proper separation, G' has fewer vertices than G. By the induction hypothesis, therefore, X is linked in G'. Replacing edges of $G' - E$ on S by paths through B as before, we can turn any linkage of X in G' into one in G, completing the proof of $(*)$. This completes the proof of (1).

Our next goal is to show that, by the induction hypothesis, we may assume that G has not only large average degree but even large minimum degree. For our proof that X is linked in G, let $s_1, \dots, s_\ell, t_1, \dots, t_\ell$ be the distinct vertices in X which we wish to link by disjoint paths $P_i = s_i \dots t_i$. Let us add to G any missing edges on X except those of

$G[X]$ the form $s_i t_i$; as the paths P_i are not allowed to have inner vertices in X, these new edges affect neither the premise nor the conclusion in $(*)$.

After this modification, we can now prove the following:

We may assume that any two adjacent vertices u, v which do not both lie in X have at least $8k - 1$ common neighbours. (2)

$e = uv$ To prove (2), let $e = uv$ be such an edge, let n denote the number of
n common neighbours of u and v, and let $G' := G/e$ be the graph obtained
G' by contracting e. Since u, v are not both in X we may view X as a subset
V' also of $V' := V(G')$, replacing u or v in X with the contracted vertex v_e if $X \cap \{u, v\} \neq \emptyset$. Our aim is to show that unless $n \geqslant 8k - 1$ as desired in (2), G' satisfies the premise of $(*)$. Then X will be linked in G' by

the induction hypothesis, so the desired paths $P_1, \dots, P_\ell$ exist in G'. If one of them contains v_e, replacing v_e by u or v or uv turns it into a path in G, completing the proof of $(*)$.

In order to show that G' satisfies the premise of $(*)$ with respect to X, let us show first that $V' \smallsetminus X$ is heavy. Since $V \smallsetminus X$ was heavy and $|V' \smallsetminus X| = |V \smallsetminus X| - 1$, it suffices to show that the contraction of e resulted in the loss of at most $8k$ edges incident with a vertex outside X. If u and v are both outside X, then the number of such edges lost is only $n+1$: one edge at every common neighbour of u and v, as well as e. But if $u \in X$, then $v \notin X$, and we lost all the X–v edges xv of G with $x \neq u$, too: while xv counted towards $\|V \smallsetminus X\|^+$, the edge xv_e lies in $G'[X]$ and does not count towards $\|V' \smallsetminus X\|^+$. If x is not a common neighbour of u and v, then this is an additional loss. But u is adjacent to every $x \in X \smallsetminus \{u\}$ except at most one (by our assumption about $G[X]$), so every such x except at most one is in fact a common neighbour of u and v. Thus in total, we lost at most $n+2$ edges. Unless $n \geqslant 8k-1$ (which would prove (2) directly for u and v), this means that we lost at most $8k$ edges, as desired for our proof that $V' \smallsetminus X$ is heavy.

It remains to show that for every small X-separation (A', B') of G' the set $B' \smallsetminus A'$ is light. Let (A', B') be a counterexample, chosen with B' minimal. Then $G'[B']$, as in the proof of (1), satisfies the premise of $(*)$ with respect to $X' := A' \cap B'$. Hence X' is linked in $G'[B']$ by induction. Let A and B be obtained from A' and B' by replacing v_e, where applicable, with both u and v, and put $X'' := A \cap B$. We shall prove that the separation (A, B) of G contradicts our assumption (1).

(A', B')

X'

(A, B), X''

Let us consider two possible positions of v_e in turn. If v_e lies in $B' \smallsetminus A'$, then $u, v \in B \smallsetminus A$. Then $X'' = X'$ is linked in $G[B]$, because it is linked in $G'[B']$: if v_e occurs on one of the linking paths for X', just replace it by u or v or uv as earlier. This contradicts (1). The other case is that v_e lies in A', possibly in X'. We show that $G[B]$ satisfies the premise of $(*)$ with respect to X''; then X'' will be linked in $G[B]$ by induction, again contradicting (1). Since (A', B') is a small X-separation, we have

$$|X''| \leqslant |X'| + 1 \leqslant |X| \leqslant 2k\,,$$

even if v_e lies in X'. Moreover, $B \smallsetminus X'' = B' \smallsetminus A'$ is heavy in G, because it is heavy in G' by the choice of (A', B'). Now consider a small X''-separation (A'', B'') of $G[B]$. Then $(A \cup A'', B'')$ is a small X-separation of G, because $|X''| \leqslant |X|$. Therefore $B'' \smallsetminus A'' = B'' \smallsetminus (A \cup A'')$ is light by the assumption in $(*)$. Hence $G[B]$ does satisfy the premise of $(*)$ for X'', completing the proof of (2).

Using induction by contracting an edge, we have just shown that the vertices in $V \smallsetminus X$ may be assumed to have large degree. Using induction by deleting an edge, we now show that their degrees cannot be too large.

d^* Since $(*)$ holds if $V = X$, we may assume that $V \setminus X \neq \emptyset$; let d^* denote the smallest degree in G of a vertex in $V \setminus X$. Let us prove the following:

$$\textit{We may assume that } 8k \leqslant d^* \leqslant 16k - 1. \tag{3}$$

The lower bound in (3) follows from (2) if we assume that G has no isolated vertex outside X, which we may clearly assume by induction. $e = uv$ For the upper bound, let us see what happens if we delete an edge e whose ends u, v are not both in X. If $G - e$ satisfies the premise of $(*)$ with respect to X, then X is linked in $G - e$ by induction, and hence in G. If not, then either $V \setminus X$ is light in $G - e$, or $G - e$ has a small X-separation (A, B) such that $B \setminus A$ is heavy. If the latter happens then e must be an $(A \setminus B)$–$(B \setminus A)$ edge: otherwise, (A, B) would be a small X-separation also of G, and $B \setminus A$ would be heavy also in G, in contradiction to our assumptions in $(*)$. But if e is such an edge then any common neighbours of u and v lie in $A \cap B$, so there are fewer than $|X| \leqslant 2k$ such neighbours. This contradicts (2).

So $V \setminus X$ must be light in $G - e$. For G, this yields

$$\|V \setminus X\|^+ \leqslant 8k\,|V \setminus X| + 1\,. \tag{4}$$

In order to show that this implies the desired upper bound for d^*, let us $f(x)$ estimate the number $f(x)$ of edges that a vertex $x \in X$ sends to $V \setminus X$. There must be at least one such edge, xy say, as otherwise $(X, V \setminus \{x\})$ would be a small X-separation of G that contradicts our assumptions in $(*)$. But then, by (2), x and y have at least $8k - 1$ common neighbours, at most $2k - 1$ of which lie in X. Hence $f(x) \geqslant 6k$. As

$$2\,\|V \setminus X\|^+ = \sum_{v \in V \setminus X} d_G(v) + \sum_{x \in X} f(x)\,,$$

an assumption of $d^* \geqslant 16k$ would thus imply that

$$2\,(8k\,|V \setminus X| + 1) \underset{(4)}{\geqslant} 2\,\|V \setminus X\|^+ \geqslant 16k\,|V \setminus X| + 6k\,|X|\,,$$

yielding the contradiction of $2 \geqslant 6k\,|X|$. This completes the proof of (3).

To complete our proof of $(*)$, pick a vertex $v_0 \in V \setminus X$ of degree d^*, and consider the subgraph H induced in G by v_0 and its neighbours. By (2) we have $\delta(H) \geqslant 8k$, and by (3) and the choice of v_0 we have $|H| \leqslant 16k$. By Lemma 3.5.4, then, H has a k-linked subgraph; let L be its vertex set. By definition of 'k-linked', we have $|L| \geqslant 2k \geqslant |X|$. If G contains $|X|$ disjoint X–L paths, then X is linked in G, as desired. If not, then G has a small X-separation (A, B) with $L \subseteq B$. If we choose (A, B) of minimum order, then $G[B]$ contains $|A \cap B|$ disjoint $(A \cap B)$–L paths by Menger's theorem (3.3.1). But then (A, B) is a linked X-separation that contradicts (1). $\square$

Exercises

For the first three exercises let G be a graph with vertices a and b, and let $X \subseteq V(G) \setminus \{a,b\}$ be an a–b separator in G.

1.$^-$ Show that X is minimal as an a–b separator if and only if every vertex in X has a neighbour in the component C_a of $G - X$ containing a, and another in the component C_b of $G - X$ containing b.

2.$^-$ (continued)

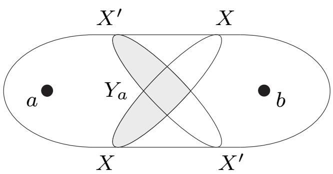
Let $X' \subseteq V(G) \setminus \{a,b\}$ be another a–b separator, and define C'_a and C'_b correspondingly. Show that both

$$Y_a := (X \cap C'_a) \cup (X \cap X') \cup (X' \cap C_a)$$

and

$$Y_b := (X \cap C'_b) \cup (X \cap X') \cup (X' \cap C_b)$$

separate a from b (see figure).



3. (continued)

Are Y_a and Y_b minimal a–b separators if X and X' are? Are $|Y_a|$ and $|Y_b|$ minimum for a–b separators if $|X|$ and $|X'|$ are?

4. Let X and X' be minimal separators in G. Show that if X meets at least two components of $G - X'$, then X' meets all the components of $G - X$ and X meets *all* the components of $G - X'$.

5.$^-$ Deduce the $k = 2$ case of Menger's theorem (3.3.1) from Proposition 3.1.1.

6.$^-$ Prove the elementary properties of blocks mentioned after their formal definition.

7. Show that the block graph of any connected graph is a tree.

8.$^-$ Let G be a k-connected graph, and let xy be an edge of G. Show that G/xy is k-connected if and only if $G - \{x,y\}$ is $(k-1)$-connected.

9. (i) Let e be an edge in a 2-connected graph $G \neq K^3$. Show that either $G - e$ or G/e is again 2-connected.

(ii) Does every 2-connected graph $G \neq K^3$ have an edge e such that G/e is still 2-connected?

10.$^+$ Let e be an edge in a 3-connected graph $G \neq K^4$. Show that either $G \dot{-} e$ or G/e is again 3-connected.